

Towards a multi-modal perceptual model

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The objective evaluation of the audio performance of nonlinear processes such as data reduction has required the development of a new generation of perception-based assessment algorithms. In the future it is expected that a large proportion of communications traffic will be wideband and multimedia. Objective assessment of perceived performance is required for optimisation during design, and for commissioning. In developing objective techniques to assess multimedia systems it is necessary to investigate, and model, the dependencies which exist between audio and visual perception (other senses may follow). This paper summarises the background of current perceptual modelling work and reports early progress towards a multisensory perceptual model. An experimental investigation into combined audio/video perception is described and an algorithmic basis for a multisensory perceptual model proposed. In the future it is predicted that algorithmic models of perception will be exploited for objective assessment of multimedia systems and by the delivery technology for advanced telepresence applications.

1. Introduction

Reliable objective assessment of perceived performance is required for optimal design, commissioning, and monitoring of quality. Nonlinear processes such as data reduction have necessitated the development of a new generation of objective assessment algorithms. Conventional engineering metrics do not take adequate account of the properties of the human receiver, and new perceptually motivated techniques have started to emerge.

Perceptually motivated audio assessment is relatively advanced and numerous models have been proposed for objectively assessing both high-quality audio, e.g. Paillard et al [1], and telephone speech quality, e.g. Hollier et al [2], and Beerends and Stemerdink [3]. Visual perceptual models which reproduce the gross psychophysics of vision are also described in the literature, usually specialising in particular aspects of visual perception, e.g. Watson and Solomon [4], and Ran and Farvadin [5]. Models of the individual senses do not reproduce the full complexity of human perception but rather allow signals to be transformed, taking account of the dominant psychophysics, to provide a more perceptually relevant representation of the signal. In this way parts of the signal which are imperceptible are given a lower weighting and parts of the signal which coincide with peaks in sensory sensitivity are given a higher weighting. The value of this approach is highlighted by the success of perceptually motivated codecs that achieve data reduction by exploiting the perceptual significance of signal components [6]. Although higher level perceptual issues have a bearing on opinions expressed by subjects the capacity to exploit basic sensory performance remains. This is the aim for multi-

sensory analysis where low-level cross-modal effects can be usefully measured and modelled with high-level cognitive issues dealt with by constraining predictions to particular application scenarios.

Advanced telepresence, virtual reality and multimedia applications and services rely on more than one of the human senses. As such services become widely used, a multisensory perceptual model will be required to account for the interaction between the senses in order to predict, and optimise, perceived performance. This paper reviews recent advances in single-sense perceptual models, reports on an early experimental investigation into cross-modal dependency, and proposes an algorithmic basis for a multisensory perceptual model.

Future trends in perceptual modelling are discussed including the need to develop more complete descriptions of the influence of task on perception. Accurate measurement of perception and the influence of task will require immersion of test subjects in highly realistic environments in order to provide algorithmic models of perception which are sufficiently robust to be exploited in the delivery technology.

2. Perceptual models

To measure the subjective performance of complex communications systems it is necessary to take account of the properties of the human senses. For example, a conventional measurement of audio signal-to-noise performance

may indicate that a system has poor performance. However, the noise component may be highly audible or quite inaudible depending on whether the noise was masked by the desired signal. A performance measurement that includes a model of the masking that occurs in hearing is capable of distinguishing between these two cases. Using models of the human senses to provide improved understanding of subjective performance is known as ‘perceptual modelling’.

2.1 Auditory perceptual models

To determine the subjective relevance of errors in audio systems, and particularly speech systems, assessment algorithms have been developed based on models of human hearing. The prediction of audible differences between a degraded and reference signal can be thought of as the sensory layer of a perceptual analysis, while the subsequent categorisation of audible errors can be thought of as the perceptual layer. Models for assessing high-quality audio, such as PERCEVAL [1], have tended only to predict the probability of detection of audible errors since any audible error is deemed to be unacceptable, while early speech models have tended to predict the presence of audible errors and then employ simple distance measures to categorise their subjective importance [2, 3, 8]. It has been previously shown [7] that a more sophisticated description of the audible error provides an improved correlation with subjective performance. In particular, the amount of error, the distribution of error, and the correlation of error with original signal have been shown to provide an improved prediction of error subjectivity.

Figure 1 shows a hypothetical fragment of an error surface. The error descriptors used to predict the subjectivity of this error are necessarily multi-dimensional — no simple single-dimensional metric can map between the error surface and the corresponding subjective opinion. The error descriptors, E_d , are in the form:

$$E_{d1} = f_{n1}\{e(i,j)\}$$

where f_{n1} is a function of the error surface element values for descriptor 1. For example, the error descriptor for the distribution of the error, error-entropy (E_e), proposed in Hollier et al [7] was given by:

$$E_e = \sum_{i=1}^n \sum_{j=1}^m a(i,j) \ln a(i,j)$$

where $a(i,j) = |e(i,j)|/E_a$, and E_a is the sum of $|e(i,j)|$ with respect to time and pitch.

$$\text{Opinion prediction} = f_{n2}\{E_{d1}, E_{d2}, \dots, E_{dn}\}$$

where f_{n2} is the mapping function between the n error descriptors and the opinion scale of interest.

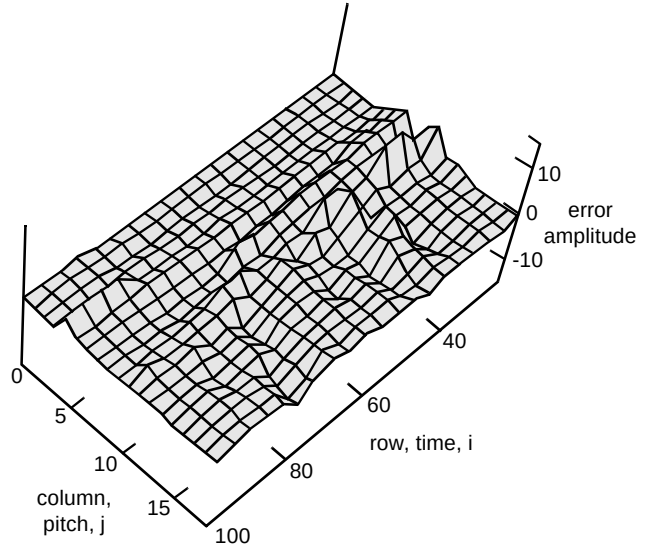


Fig 1 Fragment of audible error surface.

It has been shown that a judicious choice of error descriptors can be mapped to a number of different subjective opinion scales [9]. This is an important result since the error descriptors can be mapped to different opinion scales that are dominated by different aspects of error subjectivity. This result, together with laboratory experience, is taken to indicate that it is possible to weight a set of error descriptors to describe a range of error subjectivity since different features of the error are dominant for ‘quality’ and ‘effort’ opinion scales. The general approach of dividing the model architecture into sensory and perceptual layers and generating error descriptors that are sensitive to different aspects of error subjectivity is validated by these results.

2.2 Visual perceptual models

An approach similar to that of the auditory perceptual model has been adopted at BT Laboratories (BTL) for a visual perceptual model. A sensory layer reproduces the gross psychophysics of the sensory mechanisms:

- spatio-temporal sensitivity known as the human visual filter,
- masking due to spatial frequency, orientation and temporal frequency.

A number of visual perceptual models are known to be under development and several have been proposed in the literature. For example, Watson and Solomon [4] propose the use of Gabor functions to account for the inhibitory and excitatory influences of orientation between masker and maskee, and Ran and Farvadin [5] use a simple image decomposition into edges, textures and backgrounds. However, most of the published algorithms only succeed in optimising individual aspects of model behaviour; Watson and Solomon [4] provide a good model of masking, and Ran and Farvadin [5] a first approximation to describing the subjective importance of errors.

Following the sensory layer the image is decomposed to allow calculation of error subjectivity, by the perceptual layer, according to the importance of errors in relation to structures within the image, as shown in Fig 2.



Fig 2 Image decomposition for error subjectivity prediction.

If the visible error coincides with a critical feature of the image, such as an edge, then it is more subjectively disturbing. The basic image elements, which allow a human observer to perceive the image content, can be thought of as a set of abstracted boundaries. These boundaries can be formed by colour differences, texture changes and movement as well as edges. Even some Gestalt effects [10], which cause a boundary to be perceived, can be algorithmically predicted to allow appropriate weighting. These boundaries are required to perceive image content and so visible errors that degrade these boundaries have greater subjective significance than those which do not.

It is important to note that degradation of these boundaries can be deemed perceptually important without identifying what the high-level cognitive content of the image might be. For example, degradation of a boundary will be subjectively important regardless of whether the image is of a teapot or a spaceship. False contours in plain areas will be recognised as highly visible errors, but will not be classed as defining boundaries since they did not appear in the original image. The output from the perceptual layer is a set of context-sensitive error descriptors that can be weighted differently to map to a variety of opinion criteria.

2.3 Multisensory model

To assess a multimedia system it is necessary to combine the output from each sensory model and account for the interactions between the senses (which depend on the task or activity undertaken). It is possible to provide familiar examples of intersensory dependency, and these are useful as a starting point for discussion, despite the more sophisticated examples that soon emerge. Strong multisensory rules are already known and exploited by content providers, especially film makers. Consistent audio/video trajectories between scene cuts, and the constructive benefit of combined audio and video cues are obvious examples. Exploitation of this type of multi-modal relationship for human/computer interface design is discussed in May and Barnard [11]. Less familiar examples include the misperception of speech when audio and video cues are mismatched [12], and modification of error subjectivity with sequencing effects in the other modality [13].

The interaction between the senses can be complex and the significance of transmission errors and choice of bandwidth utilisation for multimedia services and telepresence is correspondingly difficult to determine. This difficulty highlights the need for objective measures of the perceived performance of multimedia systems. Fortunately, to produce useful engineering tools, it is not necessary to model the full extent of human perception and cognition, but rather to establish and model the gross underlying (low-level) intersensory dependencies.

Figure 3 shows a diagrammatic representation of a proposed multisensory perceptual model developing the concept of sensory and perceptual layers for the individual senses and introducing cross-modal dependencies and the influence of task.

The main components are:

- sensory models,
- cross-modal model,

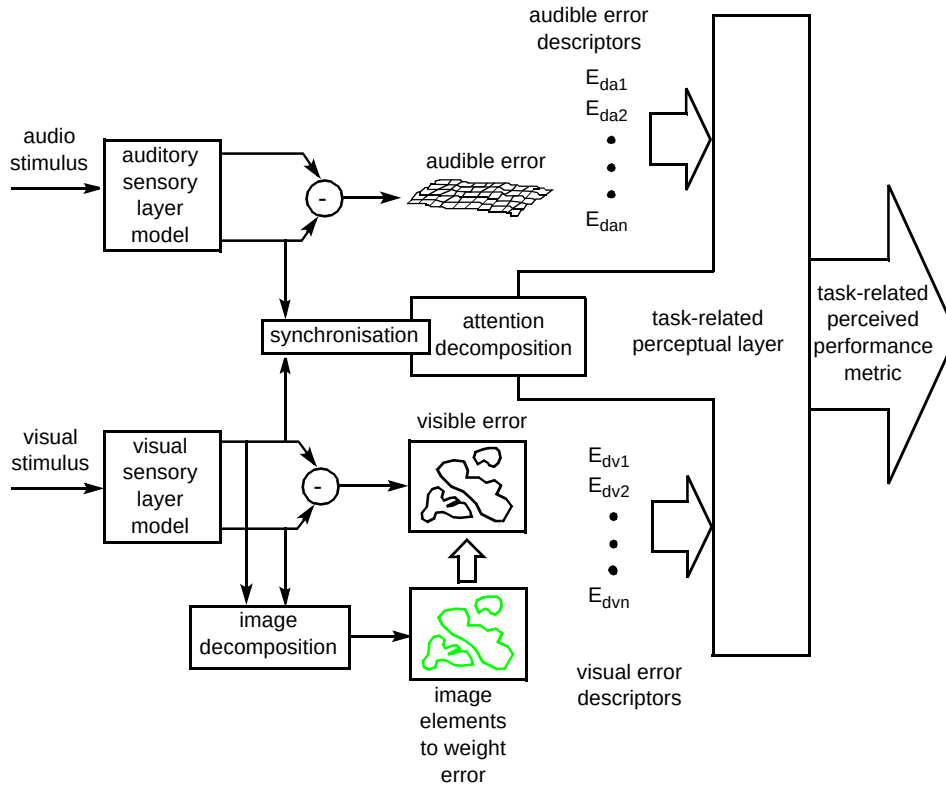


Fig 3 Overview of proposed multisensory perceptual model.

- scenario-specific task model, with an average across the target user group used for higher level perceptual issues such as personal preference and user expectation.

The main cross-modal effects are:

- timing:
 - sequencing,
 - synchronisation,
- quality balance between modalities.

Error subjectivity also depends on task:

- high-level cognitive preconceptions associated with task,
- attention split,
- degree of stress introduced by the task,
- experience of user (novice versus expert).

A mathematical structure for the model can be summarised:

$E_{da1}, E_{da2}, \dots, E_{dan}$ are the audio error descriptors, and

$E_{dv1}, E_{dv2}, \dots, E_{dvn}$ are the video error descriptors.

Then, for a given task:

fn_{aws} is the weighted function to calculate audio error,

fn_{vws} is the weighted function to calculate video error subjectivity,

fn_{pm} is the cross-modal combining function.

The task-dependent perceived performance metric, TDPM, is then:

$$TDPM = fn_{pm} \{fn_{aws} \{E_{da1}, E_{da2}, \dots, E_{dan}\}, fn_{vws} \{E_{dv1}, E_{dv2}, \dots, E_{dvn}\}\}$$

Alternatively a task term, which varies according to task, could be made explicit giving a prediction of the performance metric (PM) as:

$$PM = fn_{pm} \{fn_{aws} \{E_{da1}, E_{da2}, \dots, E_{dan}, T_{wa}\}, fn_{vws} \{E_{dv1}, E_{dv2}, \dots, E_{dvn}, T_{wv}\}\}$$

where T_{wa} is the task weighting for audio, and T_{wv} is the task weighting for video.

Proposition versus implication

The perceptual models described here rely on features that can be inferred from the audio and video signals. To weight the significance of errors, perceptual weightings must be derived from the signal without reference to higher

level cognitive knowledge of content. This approach can be referred to as an implicational model. A more complete description of error subjectivity would be possible if the higher level content was known. Use of higher level knowledge about cognitive content can be referred to as a propositional model. For example, an implicational model would predict that a high-quality image of a table with three legs contained no subjectively important errors (because the image is undegraded), while a propositional model would identify the table, noting that it should have various properties including four legs, and predict that there was an error.

The signals arriving at the current generation of perceptual models do not contain any higher level description of the content and therefore it is not possible to form and test propositions. Current work is concentrating on the extent to which perceptual importance can be deduced by implication. This approach will result in perceptual weightings being applied to regions of the image that, due to the image content or propositional considerations, are not subjectively important. However, it is reasoned that weighting all perceptually important features will provide an evaluation of the image superior to any where such a weighting has not been applied.

Future perceptual models may be able to exploit the raising and testing of propositions by utilising:

- either content descriptors supplied with the signal, for example, as proposed in MPEG7, which may be available to facilitate intelligent searches and indexing,
- or knowledge of a virtual environment within which a known range of objects and properties can exist.

The inclusion of proposition information will be useful in order to provide additional weighting on error significance. However, a deliberate attempt to develop implicational models is under way since it is expected to be difficult to exploit proposition information in certain classes of future system. For example, a normal proposition to be tested would be that an object in free space will fall. In a virtual environment this will not always be true since it would be possible, and potentially advantageous, to define some objects which remain where they are placed in space and not subject to gravity. Equally, future, advanced telepresence systems may be very unfamiliar and lead to unexpected behaviour and expectations. In such circumstances, it may be better to rely on errors which can be predicted by an implicational model rather than risk an inappropriate prediction of a propositional effect.

To develop our understanding of multi-modal perception and to provide the data required for the construction and verification of a multisensory model, it is necessary to perform a series of subjective tests. These tests

must be conducted with sufficient scientific rigour to provide hard data for numerical modelling, while at the same time be conducted efficiently so that the many tests required can be conducted with feasible resource. The technical execution of the multi-modal tests required falls beyond the scope of subjective tests previously carried out. The next section discusses a multi-modal subjective experiment, the results of this experiment, and the future requirements for multi-modal testing.

3. Experimental investigation into multi-modal perception

The first stage in the development of objective techniques to measure the subjective performance of multimedia systems is the collection of data on how these systems are perceived, and in particular the interactions which occur between the senses. To ensure that this data, which will form the basis of multi-modal models, is statistically valid, formal subjective assessments should be performed which conform to recognised and controlled design constraints. Because of the complex nature of the interactions which occur between the senses, and the influence of task and environmental factors it is necessary to ensure that subjects are immersed in highly realistic simulations of the application/task under test.

When characterising multimedia systems the potential area of investigation is vast. The scope of most early experiments is therefore to focus on specific aspects such as the influence of video quality on audio perception. These specific aspects must be explored in order to determine gross intersensory dependencies that can be used as a foundation for the initial models and as a base for further experimentation.

3.1 Report on current experiment

For this experiment, the major goal was to test and quantify the hypothesis that video quality has an influence on the perception of audio quality, as well as obtaining general information on the impact on the quality of interactions between the two. To maximise the amount of information obtained from this experiment, a non-interactive viewing-only experiment was used, as employing an interactive task (e.g. one involving human-to-human communication over a multimedia system) would have required a reduced scope to control the additional degrees of freedom. The subjects watched a set of presentations and were asked to give discrete opinions of the quality of each presentation on its completion.

Source material

The material viewed in the experiment consisted of eight video clips, each of approximately 10 s in length, with

supporting audio commentaries. The visual element consisted of a series of virtual reality fly-throughs of a hypothetical building. Unique audio commentaries were digitally recorded for each of these clips to provide two narratives one describing the software used to create the graphics and the other describing the building simulated.

One male and one female talker were used giving a total of 32 combinations (8 video clips $\times$ 2 audio themes $\times$ 2 talkers).

Degradations

The video degradations used were a sub-set of the degradations available in the ITU VIRIS software toolkit [14], as this provides algorithmically defined, well-characterised degradations. Three video degradations were selected from the toolkit:

- edge business — a distortion concentrated at the edges of objects, characterised by temporally varying sharpness or spatially varying noise, which adds a flickering type of degradation to strong edges in the image,
- blurring — makes the image blurred and unfocused,
- white noise — produces a snowstorm-like effect.

Six video degradations (three types, each at two levels), together with an undegraded condition and one with no-video gave eight video degradations.

The four audio conditions used were:

- no degradation,
- band-limited — telephone bandwidth (300—3500 Hz),
- MNRU ($Q=20$ dB) (modulated noise reference [15]¹) — this adds noise proportional to the amplitude of its input signal, is a widely used degradation in audio experiments, and is designed to simulate the effect of tandeming PCM systems,
- band-limited — a combination of the above, where the band-limiting and MNRU were performed and then the band-limited signal was passed through the MNRU.

Experiment design

To provide statistically verifiable results, a formal balanced design was used, using a pair of Graeco-Latin squares. This ensures that, over the experiment as a whole,

¹ The algorithm used was taken from the ITU Software tools library and applied directly to the commentaries sampled at 44.1 kHz. Strictly speaking, this is outside the design limits of the software, which has been designed for use with speech sampled at 8 or 16 kHz. However, it still provides an algorithmically defined degradation which was all that was required for this experiment.

all possible combinations of source material and degradations are used once and only once. In total, 32 non-expert subjects were used, each completing the experiment twice, once being asked a question on audio quality only, and once on overall quality.

Experiment procedure

The subjects were seated in an acoustically isolated room directly in front of the screen, at a distance of 5H (5 times the vertical height of the screen). A pair of speakers, one either side of the screen were used to play back the audio commentaries.

The wording for audio or overall quality was:

Please give your opinion of the overall quality of the clip.

Please give your opinion of the audio quality of the clip.

For both of these questions, the permitted answers were those used in the ITU Recommendation P.800 [16] for the quality scale:

Excellent
Good
Fair
Poor
Bad

These opinion categories are ascribed a numerical value, typically 0, 1, 2, 3, 4, which is known as an opinion score. This score may be averaged across a number of subjects to yield a mean opinion score (MOS), which is not necessarily an integer value.

Experiment results

The first stage in reviewing the results of a subjective experiment is to review their statistical validity. The balanced design methodology, permitted the use of analysis of variance (ANOVA) to estimate the experimental error and determine which of the factors within the experiment (such as subjects, degradations, presentation orders, etc) have had a significant effect on the results.

Once the analysis of variance has been completed, the scores can be averaged across the factors to yield MOSs within any constraints dictated by the ANOVA process.

Analysis of variance

An analysis of variance was performed on the results. The key points from which were:

- subject variations were significant, an expected effect, due to the random sampling from a larger population — scores can still be averaged across subjects who are considered to be a representative of the larger population,
- the order in which the subjects saw the material and the degradations did not have a significant bearing on the results,
- the variations caused by the degradations significantly affected the results,
- the question asked (audio or video quality) had a significant effect on the resulting scores,
- the ordering of the questions (whether a subject did the audio or the overall question first) also significantly affects the results,
- the material to which the degradations were applied also significantly affected the scores — this was an expected effect, as the video material was computer generated and some of the clips contained more apparent distortion artefacts than others; the use of two different talkers also accounts for some of this variation, this effect being small compared to the variations due to the degradations, and separate conclusions need not be drawn for each piece of source material.

Discussion of results

Noise was rated as the worst video degradation while the subjective responses for the other two degradations were found to be broadly similar. For the audio degradations, the order of preference (best to worst) was undegraded, band-limited, MNRU, followed by the combination of both these degradations. It is important to note that these findings do not imply that certain types of degradation are generally considered worse than others, as selecting a different severity of each of the degradations would have produced different orderings.

When no video is present, the perceived audio quality is always worse than if video is present. However, as no guidance was given to subjects on how this should be rated, interpretation of the no-video case is uncertain.

For the audio question, the effect of whether the question was asked in the first or second session has little effect on the results. However, for the overall question, the scores are higher if the overall question is asked in the first session. The notable exception to this is when no video is present, where the scores are lower.

Figure 4 summarises the effect of varying the video quality on the perception of the audio quality. The graph is averaged over all the audio qualities for clarity. The

variation in audio MOS is shown for each video distortion. To illustrate the effect of audio quality, the same data is broken down for each audio degradation in Fig 5.

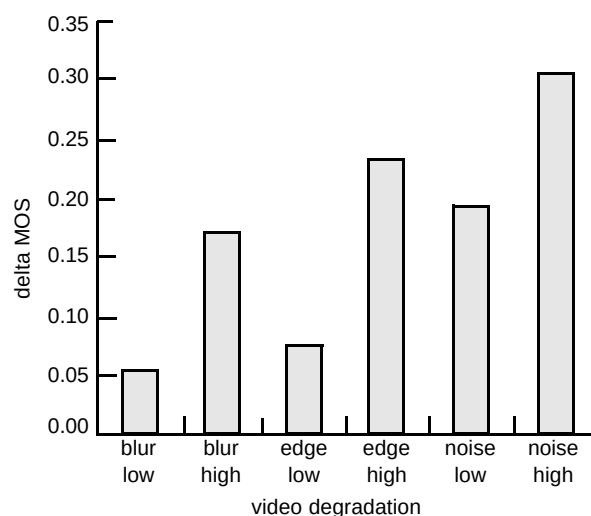


Fig 4 Differences between MOS for degraded video and for undegraded video, averaged over all audio conditions. The higher the bar, the greater this difference.

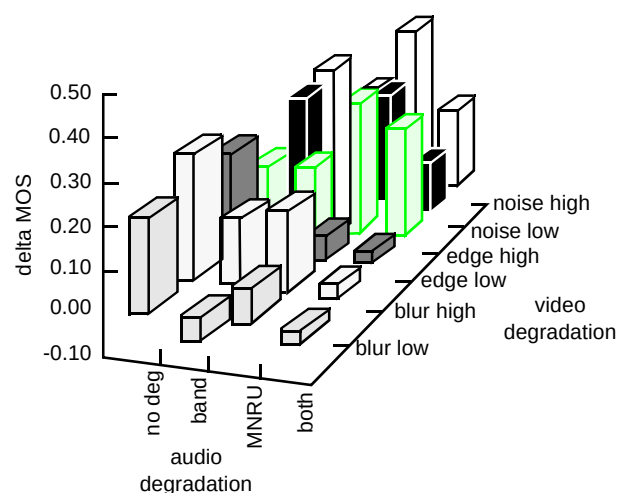


Fig 5 Differences between MOS for degraded video and for undegraded video, shown separately for each audio degradation. The higher the bar, the greater this difference.

The differences shown are small, but the trend in the data, as summarised in Fig 5, is consistent across the majority of the video degradations. It is therefore concluded that within the constraints of this experiment, a decrease in the visual quality had a detrimental effect on the perceived audio quality.

Experiment review

In the experiment reported, each opinion question (either overall or audio) was asked in separate sessions, hence the subjects knew, while viewing the clip, which

aspect of performance they would be asked to judge. This method was chosen since the object was to determine audio perception when audio was known to be the stimulus of interest. It should be noted, however, that randomising the questions and not presenting them to the subjects until they had finished viewing the clips would be expected to alter the audio/video dependency measured, since the subjects would pay attention in a more uniform way if they did not know the nature of the question.

Using a less passive, more involving task, where the subject is not able to consciously consider the audio and video performance until they are required to vote would also be likely to alter the judgements of the subjects.

Further experimentation, including questions relating to the perceived video quality, are planned to enable more information on the relationships between overall perceptions and those of the individual senses to be established.

3.2 Future multi-modal testing

It was noted above that for multi-modal perception there is a strong interdependency between perception, task and environment. Further, as with any subjective test, each user will have a variety of preconceptions, experience and aptitude. It appears impractical to separate all the potential dependencies between perception and task using a large number of individual tests and so it is proposed to immerse subjects in highly realistic simulation of the specific scenarios of interest.

If a subject is immersed in a task/application of interest it is not always practical to collect their reactions and opinions by asking explicit questions.

In order to recreate a realistic experience for test subjects, and to allow response collection without distracting from the task in hand, a new approach is required for some subjective tests. This new approach may be thought of as 'immersive subjective testing'. For example, a subject immersed in a virtual environment could be asked to perform one or more tasks. While these tasks are performed, the subject is fully involved with the task and experimental environment ensuring that preconceptions and other experiential influences are appropriately exercised. During the subject's activity within the virtual environment different aspects of performance, and preferences, can be assessed by recording reaction times and behaviour. In this way it is possible to collect a variety of multi-modal data without the explicit knowledge of the subject.

It is possible to imagine collecting data on the results of degradation of the audio/video quality, the impact of usability issues, and even factors influencing trust. During a

hypothetical information collection task a subject might encounter parts of the audiovisual environment which are degraded. The subject's reaction time and following decisions will be influenced by these distortions and using the data from many subjects the influence of the degradation, including cross-modal effects, can be determined. Direct questioning of the subject can provide the opportunity to determine the relationship between overall opinion and the responses collected during the task.

An experimental virtual environment, which was developed at BTL, is available for this type of testing and can be customised by the experimenter to suit particular requirements. A view of this experimental virtual environment is shown as Fig 7 in the next section.

4. Application of multisensory perceptual models

As sufficient information about cross-modal behaviour is collected it will be possible to build algorithmic models of multisensory perception. These models will find application in the same way that auditory perceptual models are already being exploited — for the assessment of performance, and within the delivery technology in order to reduce bandwidth and optimise performance.

4.1 Assessment

To assess the perceived performance of a multimedia system it is necessary to model both the human senses and to simulate the 'sensory interface'. The sensory interface includes the acoustic air-path between the system under test and the user, as well as the properties of the user's ear. Similarly, it is necessary to include the viewing distance between the display equipment and user's eyes. In order to assess a complete multimedia system, the most complete way to ensure that the human sensory interface is accurately simulated is to use a mannequin equipped with artificial eyes, ears and mouth. Such a system can be referred to as a 'sensory interface simulator' (SIS) and would form the front end of the multisensory perceptual model. Figure 6 shows an early prototype SIS that has been built by adding 'eye' cameras to a modified Bruel and Kjaer 4128 Head and Torso simulator. Figure 7 shows the prototype placed in an immersive interface to an experimental distributed virtual space.

4.2 Delivery technology

As multisensory models become more advanced they will be able to model cross-modal dependencies in real time and will therefore be suitable for use in the delivery technology for multimedia and advanced telepresence applications. Auditory models are already used in a similar way for coding and decoding audio signals to maximise the perceived performance of systems with limited bandwidth.

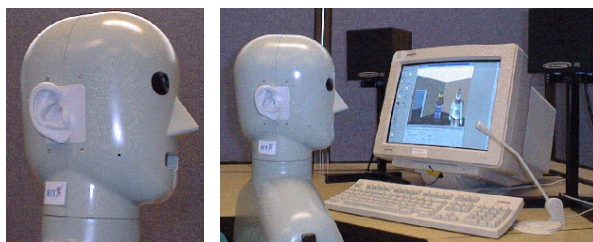


Fig 6 Close up of early prototype SIS with eyes, ears and mouth; and snap-shot of system in position in front of a distributed virtual space application.

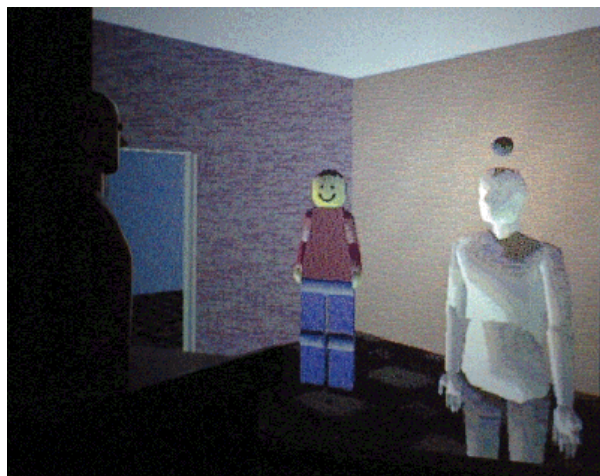


Fig 7 Early SIS prototype positioned within an immersive interface to shared virtual space.

In an advanced telepresence system there will be two principle uses for perceptual models.

- Maximise perceived performance within available bandwidth

Modelling cross-modal dependencies in real time will allow best use of available data rates, maximising the perceived performance of a multimedia system. This optimisation for data rate will be different for different users who may be joining a shared space via radically different bandwidths or by using different types of application.

- Maximise perceived performance within available processing power

Even with powerful desktop personal computers it is not possible to recreate life-like virtual spaces. The quality of the percept created can be improved by employing the available processing power to produce events and environments that are supported by cross-modal consistency. In this way, the perceived performance with the available processing power can be optimised by trading off quality audio and video according to predictions from a cross-modal model

which ensures that the required percept is effectively created. Early examples of this are likely to be relatively simple, for example:

- ‘cross-modal masking’ would allow aspects of the reproduction which will receive little attention to be rendered with less accuracy,
- ‘sequencing’, such as a temporal pattern, in one modality will be expected to increase sensitivity to the other modality in relation to the sequence.

The need to optimise multimedia systems for limited bandwidth and processing power is expected to remain an important consideration for the foreseeable future. Even when very high bandwidth connections and supercomputer processing are available to some users, it is likely that many people will wish to join virtual meetings in other ways. For example, many users will be on the move perhaps gaining access via limited radio spectrum, while the processing capabilities of portable and wearable computers is likely to be restricted for some-time. Indeed, if a compelling percept can be achieved with limited computing and bandwidth via appropriate multi-modal optimisation, this would fuel the drive towards miniaturisation rather than increased power.

5. Conclusions

The requirement for a multisensory perceptual analysis for the objective assessment of multimedia systems has been introduced. An essential part of the requirement for such an analysis is knowledge of the gross dependencies that exist between audio and video components and how to grade their impact. The important influence of higher level perceptual factors such as task and utility is also recognised.

Existing work on single-sense perceptual analysis has been summarised and attention drawn to the advantageous nature of the BTL perceptual models that provide error descriptors which may be mapped to a variety of opinion scales. Knowledge of the relationship between audio and video perception, as well as specific task-related influence on error subjectivity, is required to complete the multi-sensory model. A subjective experiment to investigate aspects of intersensory dependency has been reported and results shown which reveal a falling perception of audio quality with deteriorating video quality in certain circumstances.

A strategy has been proposed to render the overall modelling task tractable. Immersion of subjects in realistic simulations is proposed as one means to collect the required data. Gross intersensory dependency can be modelled and included in a multi-modal model with aspects of higher level task dependency included as scenario-specific weightings. An outline model structure incorporating these concepts has been shown.

The architecture of BTL's perceptual models lends itself to multisensory combination due to explicit multi-dimensional description of error subjectivity. An outline of a multisensory model is proposed and further development is planned. It is apparent that a majority of applications in the communications and entertainment industries will require multisensory evaluation.

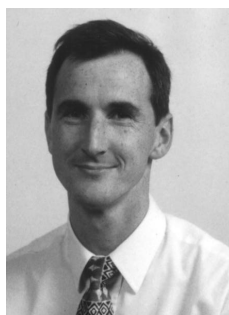
In the future, multi-modal models will be applied both for assessment and within the delivery technology for multi-media and telepresence systems.

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